

# An Exploration on the Networks of Indiana Cities and Their Structures

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**ABSTRACT.** Imagine yourself navigating through your town of residence. Do you find yourself at certain locations more often than not or have to take a very specific route to get to an area? These types of inquiries are related to how cities are structured and represented through graphs. Some places can have large amounts of connectivity while others have a few parts that act as the main bridges for accessing the rest of the network. This can lead to nodes of the city graph that serve as hubs or the main place people tend to gather. Here we discuss the top cities in Indiana in terms of population and how they're network structure compares with each other using node degree, betweenness, and KADABRA betweenness. Using these metrics, we can find the top nodes in those regards and identify them in the real world. This will showcase how impactful certain structures or locations are to these cities even if they don't appear the most assuming; these nodes carry important weight to the composition of the Indiana city networks and can reflect what residents interact with in their day-to-day lives.

## I. INTRODUCTION.

The structures of some network graphs tend to be constructed of nodes consisting of people or platforms with the edges representing relationships. For geographical networks representing cities, the roads and pathways make up the edges, but the nodes can consist of a multitude of items including buildings, landmarks, services, parks, and areas of interests among other things. As such, what can appear as a node in a city network can represent a variety of objects it can be. To learn more of these networks and their key identifiers, I decided to investigate cities from my own home state of Indiana. For years I have lived in this state and was curious as to how what aspects of it would be considered impactful in terms of network science metrics. From previous instances I have seen of Indiana city documentation, most are old articles discussing the nature of Indiana's farmland and how it has changed and developed over time [1]. Very little is mentioned about the overarching cities of the state and how they tend to be structured. To provide more information in the field of network science to a state that I have gotten familiar with, I sought to take a network science approach of the ten most populated cities in Indiana and see how they are constructed through graphs from OpenStreetMap and analysis done through Python modules to answer the questions: "How do network science measures compare and contrast highly populated Indiana cities?" and "What relationship is there with top-

measure nodes in across the different cities?" Using network science measures of degree, betweenness, and KADABRA betweenness, top nodes of each city will be found and cross-referenced with OpenStreetMap to identify what these nodes are and what they represent in the city.

## II. METHODS

To access the various network graphs of the cities, I utilized a source to help me both identify the boundaries of the city and extract the files/data containing the nodes, edges, and details of each part. Afterwards, Python code and modules would be utilized for data preparation, cleaning, analysis, and visualization.

### A. OpenStreetMap and OSMnx

OpenStreetMap is a tool created by programmers to establish a map across the world that contains map information able to be used for websites, apps, devices, etc. [2]. Within the map interface, users can view points reflecting the real-world structure of the land and extract data from portions of the map into several formats of files. Here I was able to use the Python module OSMnx which is an extension of OpenStreetMap that streamlines the process of loading locations from the map into graphs that can be utilized for Python [3]. From there I gathered the network graphs of the ten most populous cities of Indiana: Indianapolis, Fort Wayne, Evansville, South Bend, Fishers, Carmel, Bloomington, Hammond,

[1] Visher, S. S. (1966). A Brief History of Geography in Indiana. In *Proceedings of the Indiana Academy of Science* (Vol. 76, pp. 95-102).

[2] OpenStreetMap. (n.d.). *OpenStreetMap*. OpenStreetMap.

<https://www.openstreetmap.org/about>

[3] *OSMnx 1.1.2 — OSMnx 1.1.2 documentation*. (2026). Osmnx.readthedocs.io.

<https://osmnx.readthedocs.io/en/stable/>

Noblesville, and Lafayette [4]. Numbers and rankings found from this source were put into the code for future analysis. These networks would come out as directed graphs to account for directions of road travel. Next came loading each individual graph and assigning labels to their nodes and edges. To match up the information given from the city network, I also had to pull the node and edge features from OSMnx and then merge them with the nodes and edges in the city network to get a complete picture of the network information.

### B. Network Science Measure Analysis

Once the tables for the cities were prepared, I used the general network science module used for Python graphs being networkx. From there, I gathered the information from the top node regarding degree or how many total edges each node has (to remain consistent for one node per category per city, ties would go to the first node ID listed). However, there was a complication when it came to betweenness centrality or how often nodes tend to be a part of the shortest paths between other nodes in the network. Due to the large nature of the graphs, calculations can take a while to compute and run analysis. To combat this, I had the calculation observe 100 nodes in consideration for paths and have betweenness for nodes measured that way which after a few attempts appeared to give a consistent top node each time. Despite this, I decided to look at another angle to cover all of my bases and see if another measure of betweenness produced different results. For this, I used the KADABRA betweenness measure provided by the networkit module that has more measures of probability and error in place to calculate top betweenness nodes of large graphs in an efficient manner [5]. The top node in each of the three categories were used in consideration with analysis.

### C. Identification of Nodes

Lastly came the step of plugging in the top nodes into the city info tables containing the information of what type of node they are and what edges they are connected to. The information of each top node would be placed in a nested dictionary that holds the dictionary of their city's information. Columns that were considered for the node and edge information include what amenity they are, the name of the node if applicable, if they represent a building, and what's the most common path that is connected to that node.

[4] Carney, K. (2026). *Indiana Cities by Population*. [www.indiana-Demographics.com](http://www.indiana-Demographics.com). [https://www.indiana-demographics.com/cities\\_by\\_population](https://www.indiana-demographics.com/cities_by_population)

## III. RESULTS

Results of the analysis was broken down into tables and visualizations in order of the city's population going down.

TABLE I. A summary of the degree, betweenness, and KADABRA betweenness values of each city.

| City          | Degree | Between | K-Between |
|---------------|--------|---------|-----------|
| Indiana-polis | 26     | 0.2296  | 0.4906    |
| Fort Wayne    | 16     | 0.2116  | 0.4548    |
| Evansville    | 12     | 0.2550  | 0.5380    |
| South Bend    | 12     | 0.2557  | 0.4613    |
| Fishers       | 14     | 0.2543  | 0.6822    |
| Carmel        | 18     | 0.3230  | 0.8184    |
| Bloomington   | 14     | 0.1788  | 0.4118    |
| Hammond       | 12     | 0.2318  | 0.4976    |
| Noblesville   | 10     | 0.4194  | 0.8277    |
| Lafayette     | 13     | 0.1631  | 0.3839    |

Indianapolis has a high degree count which can be expected due to its high population, but cities near the middle to bottom of the list tend to have some of the higher values for both betweenness and KADABRA betweenness. Noblesville for example appears to have the highest betweenness in both categories with Carmel being second in both. Smaller and larger cities tend to have similar values of betweenness overall, so this can showcase how cities such as Noblesville and Carmel are less connected than others and have locations that the population more often than not interact with.

[5] *Class KadabraBetweenness*. (2016). Github.io. [https://networkit.github.io/dev/docs/cpp\\_api/classNetworkKit\\_1\\_1KadabraBetweenness.html](https://networkit.github.io/dev/docs/cpp_api/classNetworkKit_1_1KadabraBetweenness.html)

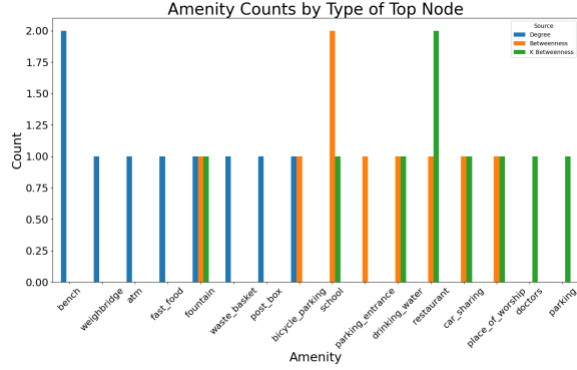


FIG 1. A grouped bar chart highlighting the different amenity counts per category of node (blue: degree, orange: betweenness, green: KADABRA betweenness)

Looking at the different amenity counts for the different types of nodes in Figure 1, there's an interesting distinction that appears. The top counts for the two betweenness measures show schools and restaurants along with other buildings placed around such as places of worship and a doctor's office. However, what's present in all the categories and most prominent in the degree categories are unassuming features of cities such as benches, parking areas, and fountains. These areas represent more so public areas where people can freely interact with these nodes whether that be using the feature or simply passing by it along a path.

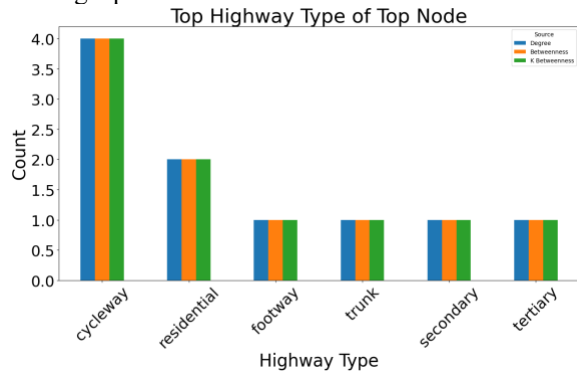


FIG 2. A grouped bar chart highlighting the different highway counts that were the most common for a node's connections per category of node (blue: degree, orange: betweenness, green: KADABRA betweenness)

Figure 2 represents the counts of the most common type of highway or path for each of the nodes and there appears to be a similar idea present for the top three types; Cycleways and footways demonstrate people accessing these top nodes through alternate methods while residential indicates roads through neighborhoods and other similar paths with houses. These three types represent smaller paths away from

the typical idea of highways of large stretches of roads with lanes of vehicles. This appears to show that most frequented nodes in a city are tied into areas people can pass by and can access in a simpler manner than having to drive there. This falls in line with the numbers in Figure 1 where more public areas tend to be the most impactful nodes when it comes to Indiana city networks.

#### IV. DISCUSSION

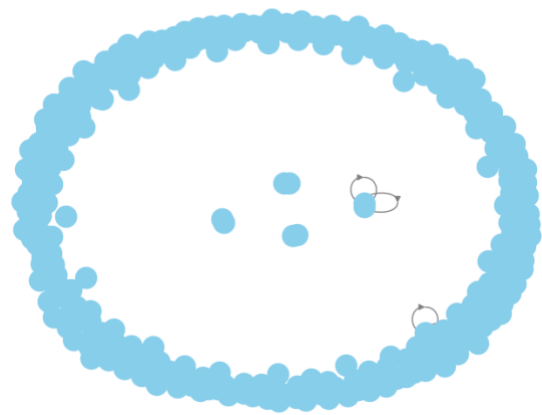
Regarding the differences in city structure across Indiana, it seems that their top degree nodes vary according to their population with Indianapolis having a substantial lead over the rest. However, the highest betweenness appears to have more variance present with Carmel and especially Noblesville having higher values. For KADABRA betweenness, the values tend to be more standardized with higher values having less of a lead and all values having the same rank as they did in the betweenness category. Regardless, the same rankings show Carmel and Noblesville have more impactful nodes that connect the city which could come down to city structure and having less options to traverse different areas. This in turn could cause more traffic in the city especially if the high betweenness node represents a popular area. For amenities tied to the top nodes, it seems that there's a trend of public areas occupying more of the impactful nodes rather than features of importance in the city such as strong businesses or landmarks. This idea can represent that the average resident must frequent the area to get where they need to go which can imply that accessible paths such as cycleways and footways make up a large portion of a resident's paths for travel. As such, top nodes of networks of cities in Indiana appear to have more emphasis on simpler pathways that aren't encumbered by multi-lane roads but rather focus on the public areas that all residents frequent and interact with as part of society. These areas should be well maintained to ensure that any residents and visitors can navigate the city with ease and keep the network flowing without bottlenecks or issues.

#### V. CONCLUSIONS

Overall, the most populated cities in Indiana follow a general trend when it comes to the layout of central network science measures and identity of top nodes in said measures. Despite Indianapolis having a fairly large population and degree compared to the rest of the cities, its top measure in betweenness was fairly standard compared to the rest with Carmel and Noblesville having a larger top node in betweenness and KADABRA betweenness representing more limited options for traversing the city. The emergence of public spaces as the predominant feature in the top

nodes signifies the importance of open areas in Indiana cities where people will be more often than not utilizing these areas for movement, an opportunity to leave their vehicle in a parking area to achieve errands, or perhaps to spend some time amidst their day-to-day lives. I believe that these areas provide large importance to the city as there's less linearity in these pathways compared to a straight road and there could be more options available to travel such as with a bike or a skateboard. While a car may be necessary to move around efficiently in cities such as those in Indiana, having connections such as these walkable areas can provide more opportunities to residents to move around and ultimately reduce traffic in the network and helps people become more efficient when navigating their city.

Random Subset of Bloomington City Network

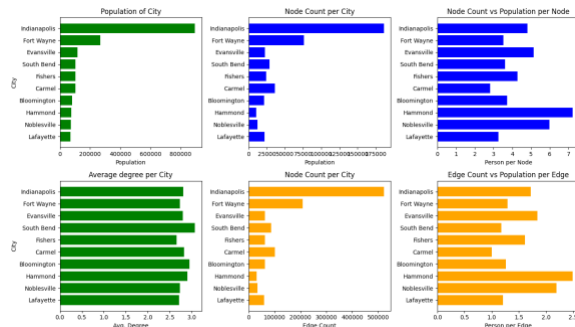


An example of a subset of nodes in the Bloomington network (limited due to the high number of nodes).

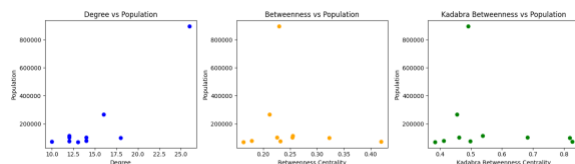
## APPENDIX A: GitHub

Access to GitHub for files and project work:  
<https://github.iu.edu/ebmontoy/INFO-I606>

## APPENDIX B: Estra Visualizations



Statistical summary of the different cities



Scatterplots featuring top degree, betweenness, and KADABRA betweenness nodes versus population.